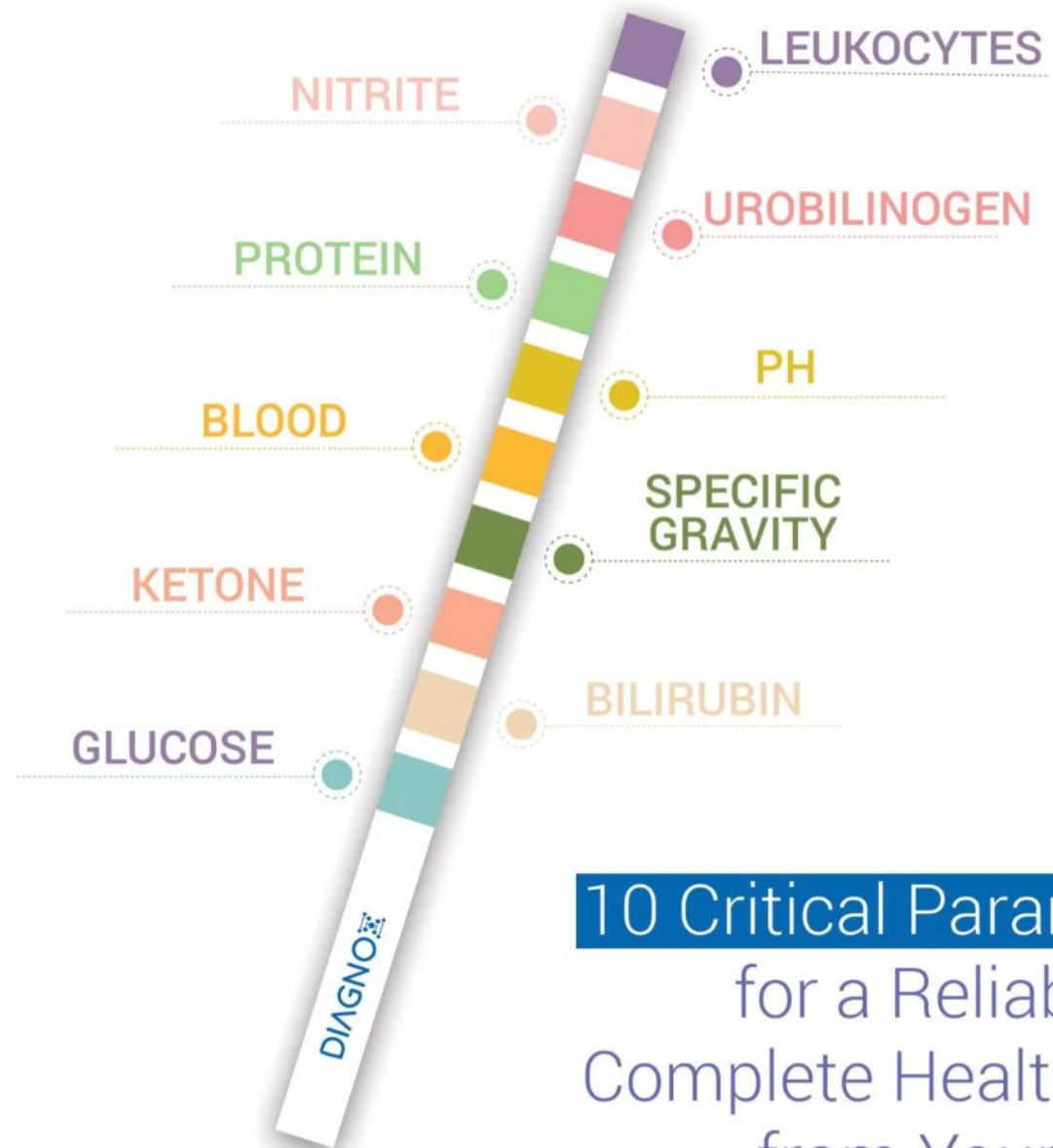


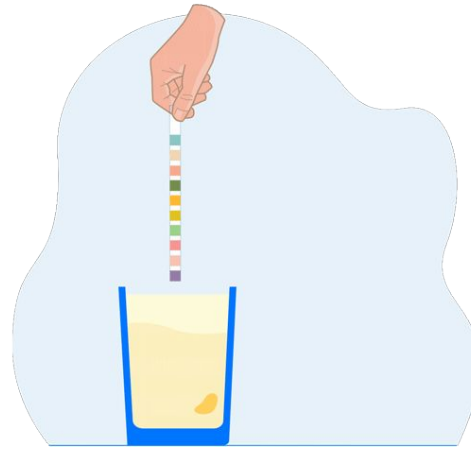
Smartphone-Based Point of Care Urinalysis Assessment



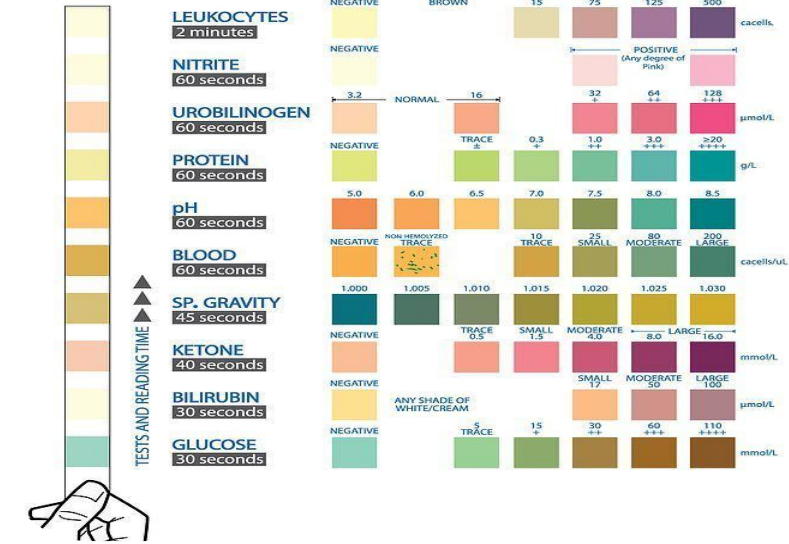
Traditional Procedure



Extract test strip.



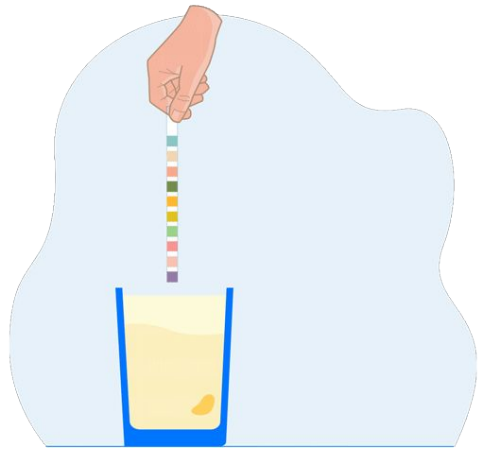
Immerse in specimen.



Manually match colors.

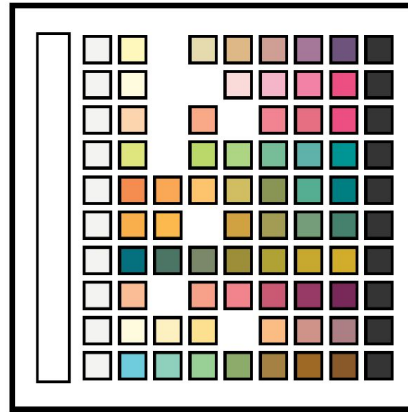
Overview of Automated Procedure

Step 1



Immerse
test strip.

Step 2



Place it on
chart.

Step 3



Get image &
view results

Algorithm Overview

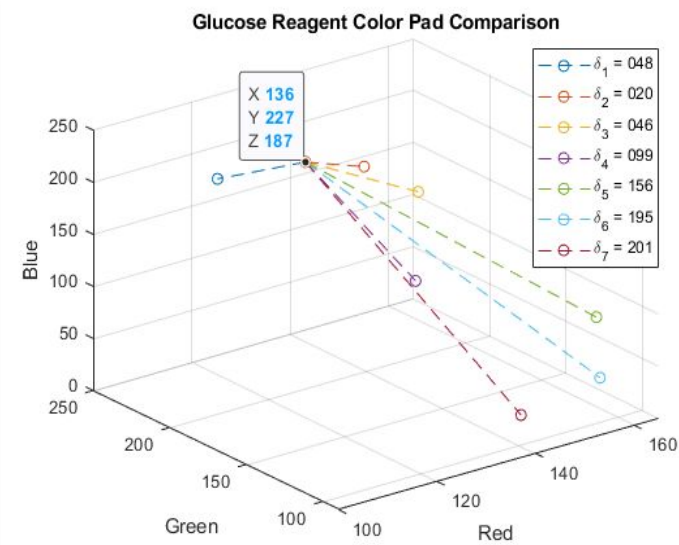
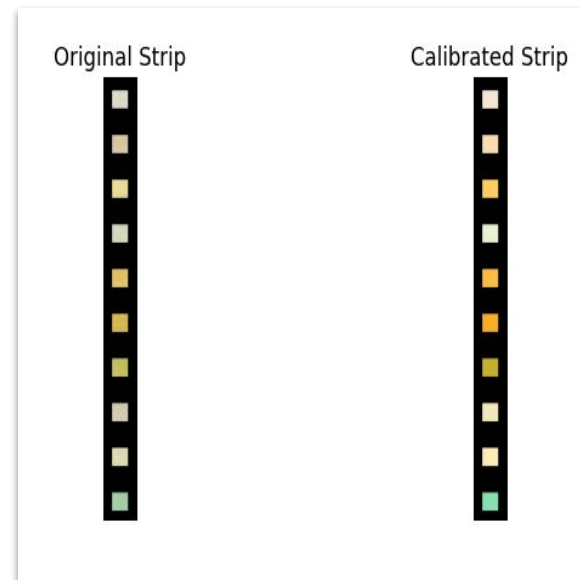
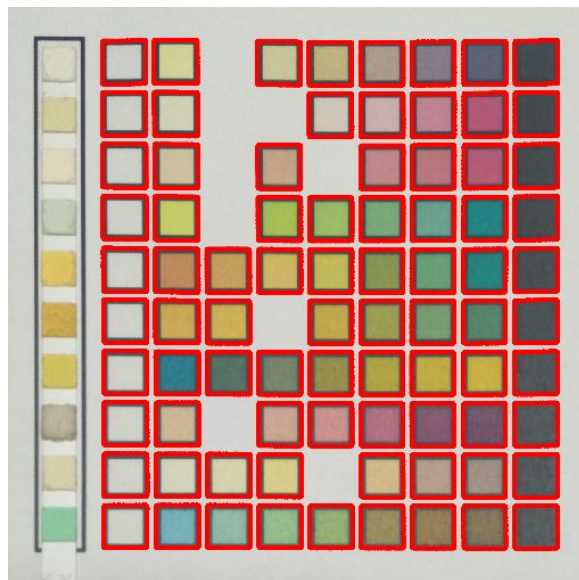
1. Calibration Chart Design.

2. Segmentation of color swatches.

3. Light and color calibration of test strip.

4. Comparison with reference chart and report test results.

Algorithm Steps



Algorithm Advantages



Accurate results under different lightening conditions.

Consistent performance across different smart phone cameras.

Economic & ubiquitous solution

Greater connectivity & medical record keeping.

Application for In-House Testing

CARE [Urinox Project]

In-House Testing Application

Patient Information

Name Age Gender Date & Time

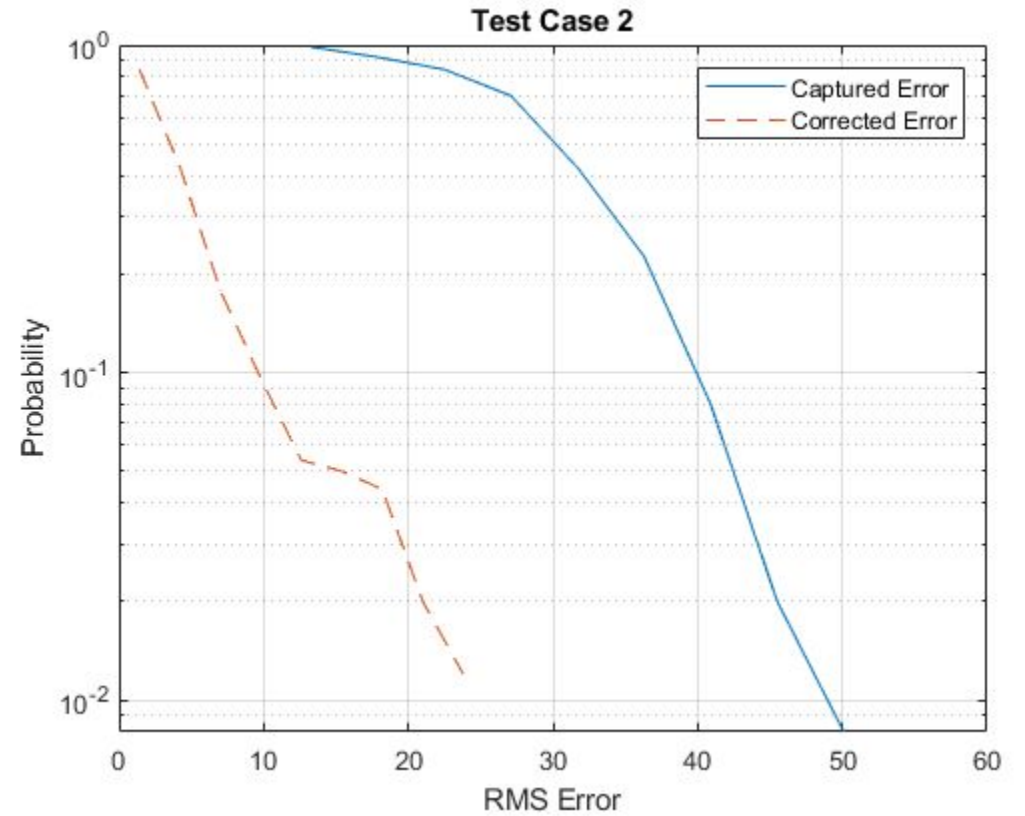
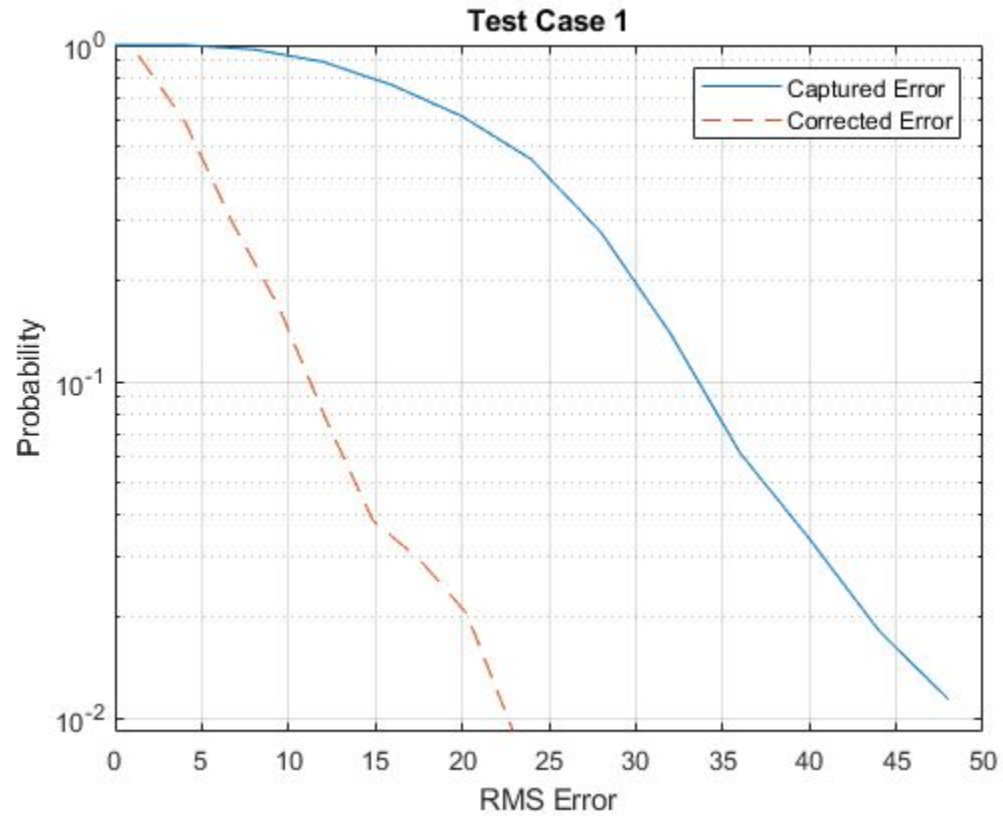
Image File

Image Preview

Test Results

Leukocytes	
Nitrite	
Urobilinogen	
Protein	
pH	
Blood	
Sp. Gravity	
Ketone	
Bilirubin	
Glucose	

Performance across Smartphones



Thank you!

